

Impact craters

Our planet is continually showered by meteorites. Most of these are very small, but sometimes Earth is struck by a meteorite big enough to create an impact crater, just like the ones you can see on the Moon. It is likely that at least one such impact caused a mass extinction of life long ago. Simple craters are deep, but complex craters are wider, often with rings of ridges and a central dome.

Barringer Crater

United States



Location Painted Desert, Arizona

Date of impact 50,000 years ago

Width 0.75 miles (1.2 km)

Type Simple

Also known as Meteor Crater, this is one of the best-preserved impact sites on the planet. It was probably made by a nickel-iron meteorite roughly 150 ft (50 m) across. Remains of the meteorite are still scattered around the rim of the crater, in

the form of tiny metal balls that formed when it vaporized on impact.

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Barringer Crater was caused by a recent impact, which is why it is still well defined.



Spider Crater

Australia



Location Kimberley Plateau,

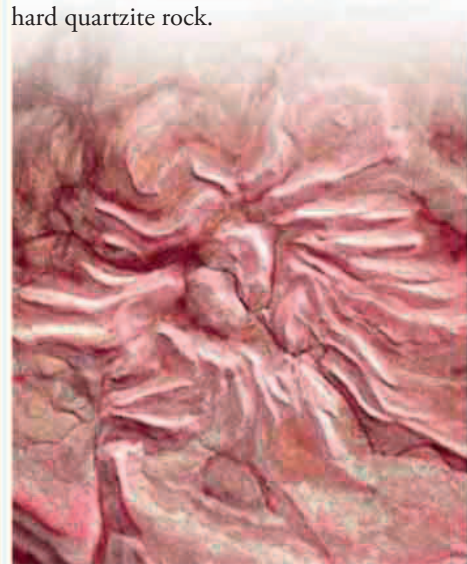
Western Australia

Date of impact 600–900 million years ago

Width 8 miles (13 km)

Type Complex

Hidden away in a remote region of northwest Australia, this very ancient structure was only identified in the 1970s. The crater has a massively eroded central dome created by the recoil of Earth's crust after the impact. This is now carved into spiderlike, radiating ridges of very hard quartzite rock.



Lonar Crater

India



Location Maharashtra, near Mumbai

Date of impact 50,000 years ago

Width 1.1 miles (1.8 km)

Type Simple

Lonar Crater is similar to Barringer Crater, but slightly bigger. It was formed at about the same time in very hard basalt rock, and is beautifully preserved. The monsoon climate has turned it into a lake 560 ft (170 m) deep. It is almost perfectly circular and surrounded by dense woodland and several Hindu temples.